

HYDROGENATED NBR (HNBR) PREPARATION PROCESSES: CATALYST-ENABLED SIMULTANEOUS HYDROGENATION AND METATHESIS

1. ABSTRACT

This report synthesizes two primary patents focused on the preparation of hydrogenated nitrile rubber (HNBR) from nitrile rubber precursors. The target material is hydrogenated or partially hydrogenated NBR with controlled residual unsaturation, improved heat/oil/oxidative resistance, and reduced molecular weight or narrower molecular-weight distribution relative to conventional HNBR. Both patents address the same core transformation: selective hydrogenation of acrylonitrile-butadiene rubber in the presence of hydrogen, but they emphasize catalyst-enabled process intensification rather than conventional stepwise hydrogenation. EP2473280B1 discloses a simultaneous hydrogenation and metathesis route using ruthenium- or osmium-based catalysts of general formulae (I)–(III), with examples comparing Grubbs II, Grubbs III, and Wilkinson's catalyst. The patent highlights faster reduction of residual double bonds and the ability to obtain HNBR with lower molecular weight and narrower distribution in a single step. US9150669B2 similarly discloses a simultaneous hydrogenation/metathesis process, but focuses on alternative catalyst systems of formula (I) and explicitly frames the invention as a route to low-molecular-weight HNBR with low residual double bond content. Across the landscape, the dominant process trend is catalytic hydrogenation coupled to metathesis, using ruthenium-based complexes and optional co-olefin to tune chain scission and molecular weight. The disclosed examples emphasize catalyst identity, catalyst loading in phr, metal content in ppm, reaction time, and residual double-bond conversion profiles as the key process variables. Overall, the patents indicate that HNBR synthesis can be improved by selecting metathesis-active hydrogenation catalysts that simultaneously reduce unsaturation and control molecular weight, enabling a single-step preparation of HNBR with targeted viscosity and performance properties.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: EP2473280B1

- Patent Title: Process for the preparation of hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a one-step process for producing hydrogenated nitrile rubber by simultaneously hydrogenating and metathesizing nitrile rubber in the presence of hydrogen, optionally with a co-olefin. The process uses catalysts of general formulae (I), (II), or (III), centered on ruthenium or osmium complexes, to achieve both backbone saturation and molecular-weight reduction. - Hydrogenation catalyst: Grubbs III — Example 1 - Hydrogenation catalyst: Grubbs III — Example 1 - Hydrogenation catalyst: Wilkinson's — Example 1 - Hydrogenation catalyst: Grubbs II — Example 1 - Catalyst amount: 0.057 phr — Example 1, HNBR 1 - Catalyst amount: 0.080 phr — Example 1, HNBR 2 - Catalyst amount: 0.06 phr — Example 1, HNBR 3 - Catalyst amount: 0.055 phr — Example 1, HNBR 4 - Metal content: 65.6 ppm — Example 1, HNBR 1 - Metal content: 91.3 ppm — Example 1, HNBR 2 - Metal content: 65.6 ppm — Example 1, HNBR 3 - Metal content: 65.6 ppm — Example 1, HNBR 4 - Reaction time: 0 min — Example 1, Table 2 - Residual double bond content (RDB): 100% — Example 1, Table 2, all HNBR samples at 0 min - Reaction time: 60 min — Example 1, Table 2 - RDB: 2.0% — Example 1, HNBR 1 at 60 min - RDB: 0.7% — Example 1, HNBR 2 at 60 min - RDB: 15.4% — Example 1, HNBR 3 at 60 min - RDB: 2.9% — Example 1, HNBR 4 at 60 min - Reaction time: 120 min — Example 1, Table 2 - RDB: 0.4% — Example 1, HNBR 1 at 120 min - RDB: 0.5% — Example 1, HNBR 2 at 120 min - RDB: 3.1% — Example 1, HNBR 3 at 120 min - RDB: 0.6% — Example 1, HNBR 4 at 120 min - Reaction time: 180 min — Example 1, Table 2 - RDB: 0.3% — Example 1, HNBR 1 at 180 min - RDB: 0.3% — Example 1, HNBR 2 at 180 min - RDB: 1.3% — Example 1, HNBR 3 at 180 min - RDB: 0.3% — Example 1, HNBR 4 at 180 min - Reaction time: 240 min — Example 1, Table 2 - RDB: 0.6% — Example 1, HNBR 3 at 240 min - RDB: 191541 g/mol — Example 1, HNBR 1 Mw - Mn: 68643 g/mol — Example 1, HNBR 1 - PDI: 2.79 — Example 1, HNBR 1 - Mooney viscosity: 38.2 — Example 1, HNBR 1 - Mw: 135772 g/mol — Example 1, HNBR 2 - Mn: 58259 g/mol — Example 1, HNBR 2 - PDI: 2.33 — Example 1, HNBR 2 - Mooney viscosity: 20.2 — Example 1, HNBR 2 - Mw: 195304 g/mol — Example 1, HNBR 3 - Mn: 64130 g/mol — Example 1, HNBR 3 - PDI: 3.05 — Example 1, HNBR 3 - Mooney viscosity: 69.3 — Example 1, HNBR 3 - Mw: 144212 g/mol — Example 1, HNBR 4 - Mn: 59851 g/mol — Example 1, HNBR 4 - PDI: 2.45 — Example 1, HNBR 4 - Mooney viscosity: 32.6 — Example 1, HNBR 4

Relevant Experimental Evidence - Example 1: Compared Grubbs III, Wilkinson's, and Grubbs II catalysts for simultaneous hydrogenation/metathesis and showed faster early-stage RDB reduction with Grubbs III. - Example 1: Demonstrated that HNBR samples reached very low residual double bond content after 120-180 min, depending on catalyst. - Example 1: Showed that catalyst choice affected molecular weight and Mooney viscosity, with HNBR 2 giving the lowest

Mw among the listed samples.

Technical Relevance This patent is directly relevant because it targets HNBR formation from nitrile rubber using catalytic hydrogenation, while also controlling molecular weight through metathesis. It provides concrete comparative evidence that catalyst selection strongly influences hydrogenation rate, residual unsaturation, and final polymer rheology.

Patent 2

- Patent Number: US9150669B2
- Patent Title: Process for the preparation of low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a process for producing low-molecular-weight hydrogenated nitrile rubber by contacting nitrile rubber with a ruthenium- or osmium-based catalyst in the presence of hydrogen, optionally with a co-olefin. The process is presented as a simultaneous hydrogenation and metathesis route designed to lower residual double bond content while reducing molecular weight and narrowing molecular-weight distribution. - Hydrogenation catalyst: Hoveyda — Example 1 - Hydrogenation catalyst: Arlt — Example 1 - Hydrogenation catalyst: Wilkinson's — Example 1 - Catalyst amount: 0.041 phr — Example 1, HNBR 1 - Catalyst amount: 0.045 phr — Example 1, HNBR 2 - Catalyst amount: 0.060 phr — Example 1, HNBR 3 - Metal content: 65.6 ppm — Example 1, HNBR 1 - Metal content: 65.6 ppm — Example 1, HNBR 2 - Metal content: 65.6 ppm — Example 1, HNBR 3 - Co-catalyst: NIL — Example 1, HNBR 1 - Co-catalyst: NIL — Example 1, HNBR 2 - Co-catalyst: Triphenylphosphine — Example 1, HNBR 3 - Co-catalyst amount: 1.0 phr — Example 1, HNBR 3 - Reaction time: 0 min — Example 1, Table 2 - Residual double bond content (RDB): 100% — Example 1, Table 2, all HNBR samples at 0 min - Reaction time: 60 min — Example 1, Table 2 - RDB: 2.5% — Example 1, HNBR 1 at 60 min - RDB: 7.0% — Example 1, HNBR 2 at 60 min - RDB: 15.4% — Example 1, HNBR 3 at 60 min - Reaction time: 120 min — Example 1, Table 2 - RDB: 0.6% — Example 1, HNBR 1 at 120 min - RDB: 4.5% — Example 1, HNBR 2 at 120 min - RDB: 3.1% — Example 1, HNBR 3 at 120 min - Reaction time: 180 min — Example 1, Table 2 - RDB: 0.5% — Example 1, HNBR 1 at 180 min - RDB: 3.9% — Example 1, HNBR 2 at 180 min - RDB: 1.3% — Example 1, HNBR 3 at 180 min - Reaction time: 240 min — Example 1, Table 2 - RDB: 0.4% — Example 1, HNBR 1 at 240 min - RDB: 3.5% — Example 1, HNBR 2 at 240 min - RDB: 0.7% — Example 1, HNBR 3 at 240 min - Mw: 113371 g/mol — Example 1, HNBR 1 - Mn: 48352 g/mol — Example 1, HNBR 1 - PDI: 2.34 — Example 1, HNBR 1 - Mooney viscosity: 33.3 — Example 1, HNBR 1 - Mw: 133758 g/mol — Example 1, HNBR 2 - Mn: 54595 g/mol — Example 1, HNBR 2 - PDI: 2.45 — Example 1, HNBR 2 - Mooney viscosity: 41.3 — Example 1, HNBR 2 - Mw: 195304 g/mol — Example 1, HNBR 3 - Mn: 64130 g/mol — Example 1, HNBR 3 - PDI: 3.05

— Example 1, HNBR 3 - Mooney viscosity: 69.3 — Example 1, HNBR 3

Relevant Experimental Evidence - Example 1: Showed that the Hoveyda-type catalyst (HNBR 1) gave the fastest hydrogenation and the lowest final RDB among the compared systems. - Example 1: Demonstrated that the catalyst system reduced molecular weight substantially, with HNBR 1 and HNBR 2 both below the industrial reference HNBR 3. - Example 1: Showed that adding triphenylphosphine as co-catalyst in HNBR 3 did not outperform the inventive catalyst systems in hydrogenation efficiency.

Technical Relevance This patent is highly relevant because it provides a practical catalyst-screening framework for producing low-molecular-weight HNBR in a single step. It directly links catalyst identity and co-catalyst choice to residual unsaturation, molecular weight, and viscosity, which are central control variables in HNBR synthesis.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- Both patents use simultaneous hydrogenation and metathesis of nitrile rubber as the core HNBR-forming strategy, rather than conventional post-hydrogenation alone.
- EP2473280B1 emphasizes Grubbs III, Grubbs II, and Wilkinson's catalyst comparisons, while US9150669B2 emphasizes Hoveyda, Arlt, and Wilkinson's catalyst systems.
- Catalyst loadings are reported in phr in both patents, but EP2473280B1 spans 0.055-0.080 phr and US9150669B2 spans 0.041-0.060 phr in the main example.
- Both patents report 65.6 ppm metal content for most samples; EP2473280B1 also includes a 91.3 ppm case, whereas US9150669B2 keeps all compared samples at 65.6 ppm.
- EP2473280B1 shows very rapid RDB reduction to 0.4-0.5% by 120 min for the best catalysts, while US9150669B2 shows 0.6% at 120 min for HNBR 1 and slower conversion for HNBR 2 and HNBR 3.
- US9150669B2 achieves lower Mw for the best sample (113371 g/mol) than EP2473280B1's best listed sample (135772 g/mol), indicating stronger molecular-weight reduction under its catalyst set.
- Both patents support the same process trend: ruthenium-based metathesis-active hydrogenation catalysts can simultaneously lower unsaturation and molecular weight to produce HNBR with improved processability.

4. CONCLUSION

The evidence supports a catalyst-driven, single-step route to HNBR from nitrile rubber using simultaneous hydrogenation and metathesis. The most effective disclosed conditions combine ruthenium-based catalysts with low phr loadings, metal contents around 65.6 ppm, and reaction times of 120-240 min, yielding residual double bond contents near or below 1% and substantially

reduced molecular weight. Based on the examples, the most favorable synthesis window is a metathesis-active hydrogenation catalyst system that balances rapid unsaturation removal with controlled chain scission, with EP2473280B1 favoring Grubbs III-type systems and US9150669B2 favoring Hoveyda-type systems. For HNBR synthesis, the evidence recommends prioritizing ruthenium catalyst selection, maintaining low catalyst loading, and monitoring RDB and Mw as the key endpoints for process optimization.

5. REFERENCES

- EP2473280B1 | Process for the preparation of hydrogenated nitrile rubber | Not disclosed | EP | 2014 | <https://patents.google.com/patent/EP2473280B1/en>
- US9150669B2 | Process for the preparation of low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2015 | <https://patents.google.com/patent/US9150669B2/en>